STM Exercise 3

- Creep
- Fatigue

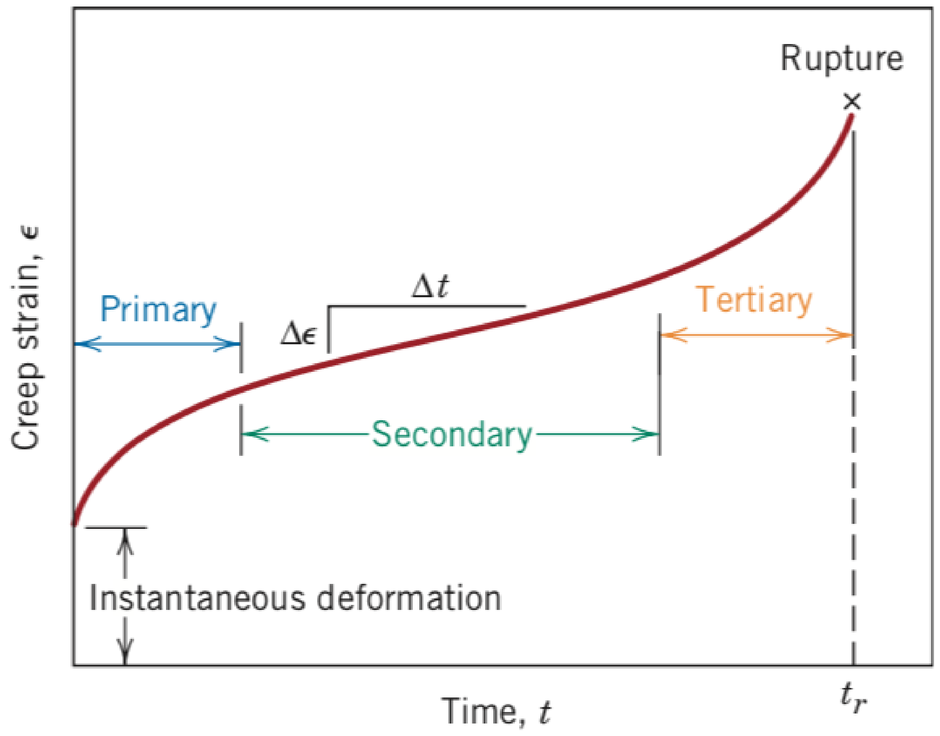
CREEP

creep test: a constant load is applied at a given T, strain is recorded versus time.

Constant load in the elastic field can give to a plastic deformation, progressive to fracture.

1. Thermally activated process (High T Process)

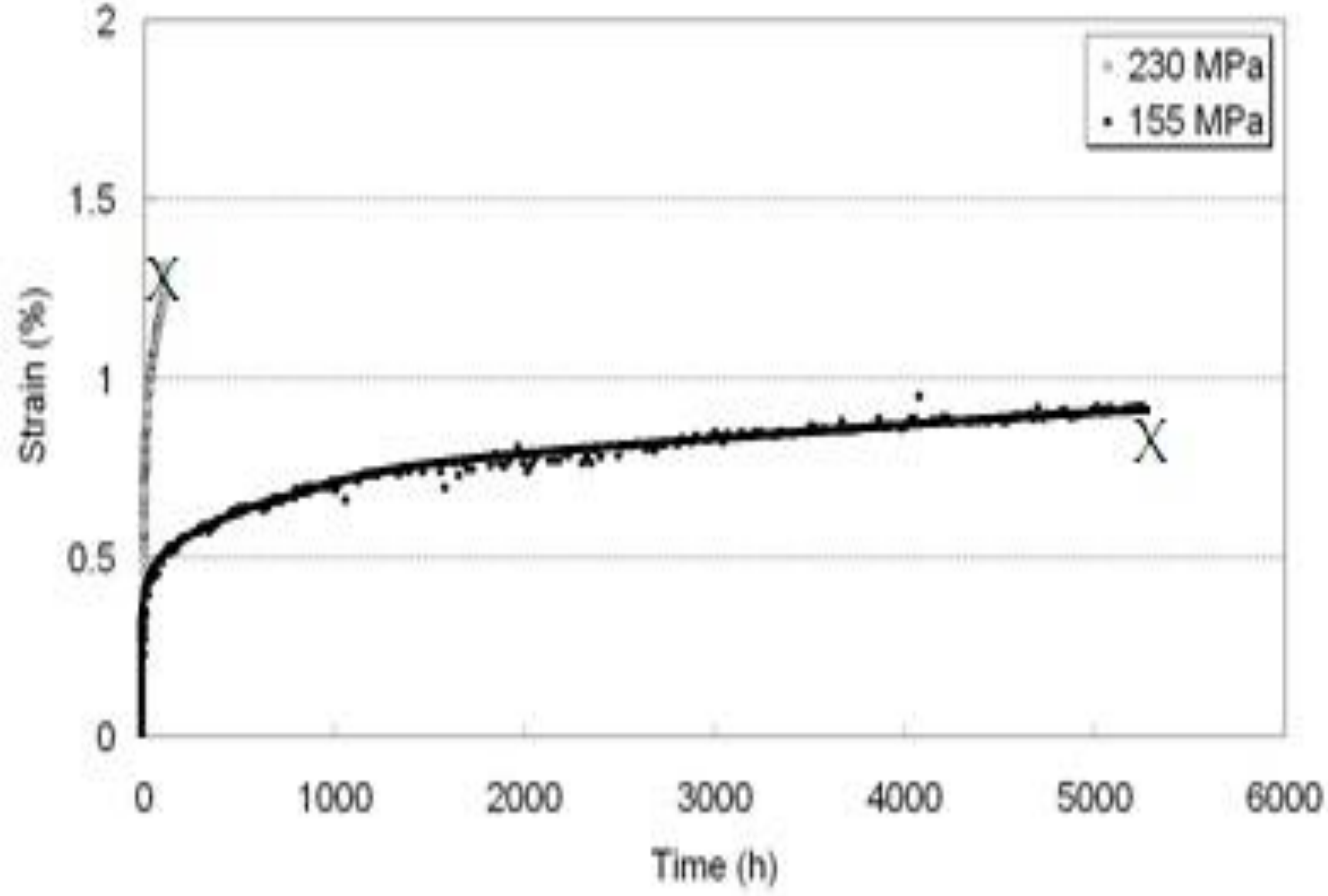
Metals **T>0,3-0,4 T$_{\mathbf{melting}}$**
Ceramics **T> 0,6-0,7 T$_{\mathbf{melting}}$**
Amorphous materials **T>T$_{\mathbf{glass}}$**



1- Draw tensile and creep curves of a given metal with the following microstructures and Briefly explain the differences: mono-crystalline, polycrystalline with large grains, polycrystalline with small grains.

2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at T1>T2. Explain differences.

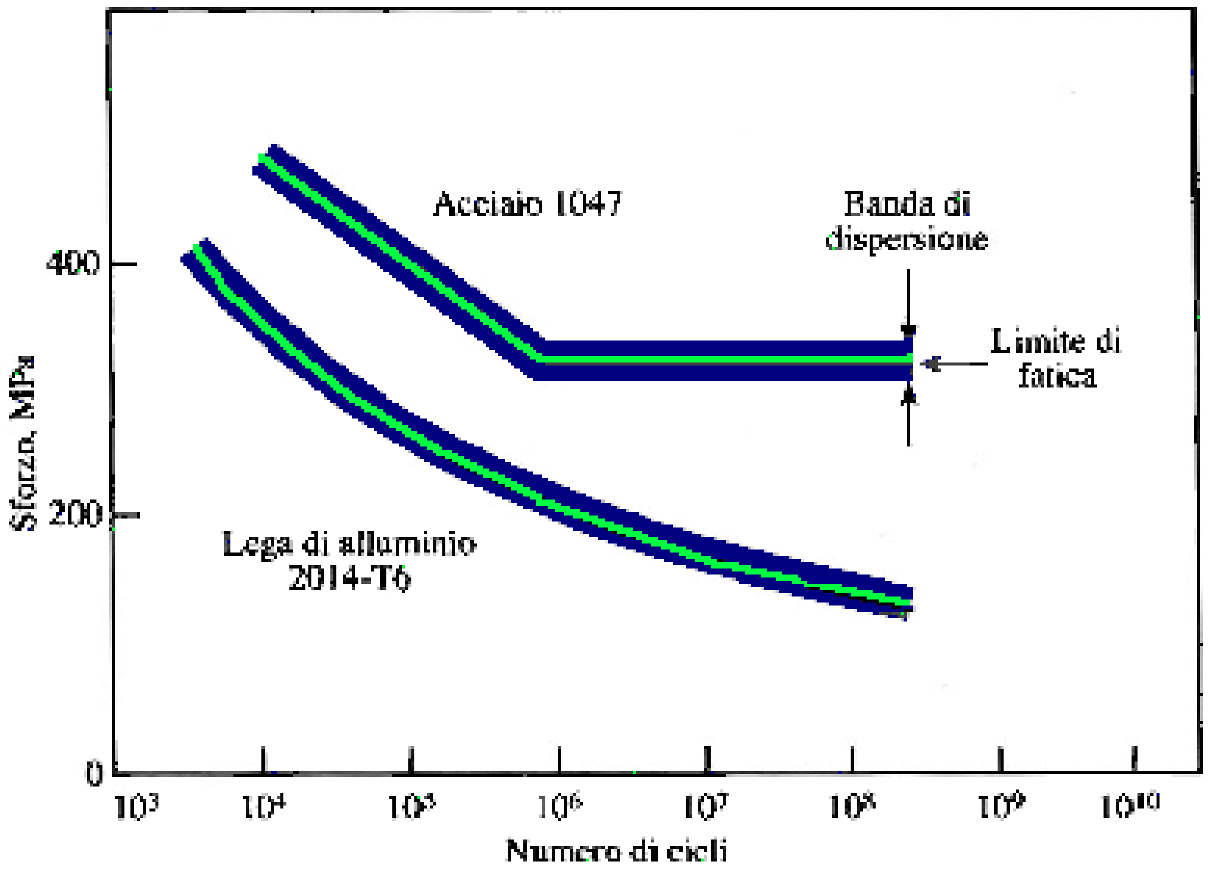
3- Indicate on the creep curve below the plastic and elastic deformation of a sample loaded at stress=155 MPa after 3000 hours.



4- Draw the stress/strain curves after tensile tests of a metal at room temperature (RT) and at T1 > RT. Briefly explain the difference.

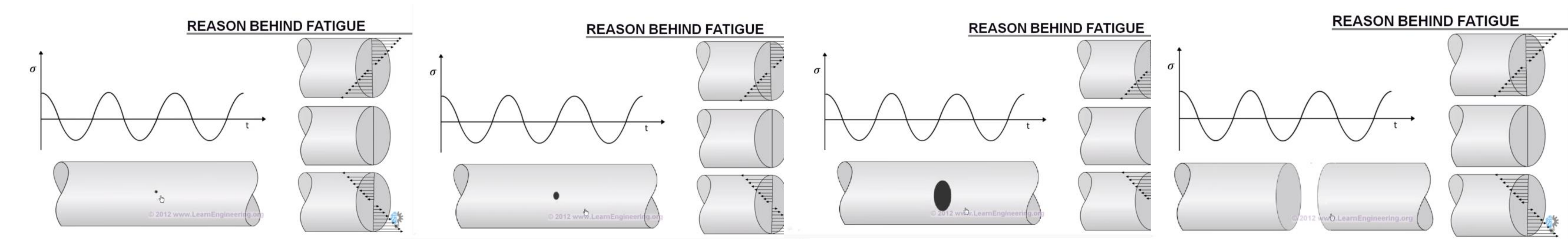
Fatigue

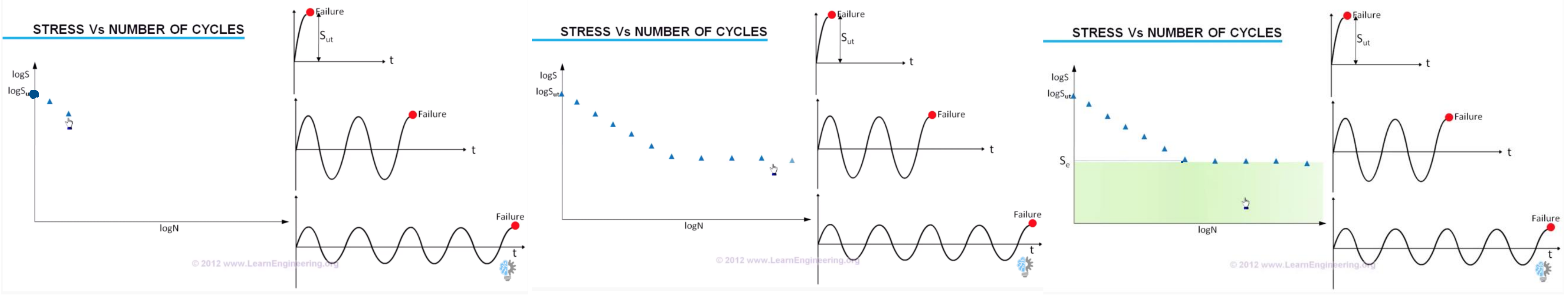
- Failure of materials due to cyclic loading.
- Main reason of mechanical failure of materials
- Failure happens at stress lower than σ_R or σ_Y
- Catastrophic failure of materials (also for ductile materials !)
- Fatigue tests: materials are cyclically loaded at different stresses up to failure.
- Fatigue limit: when cyclically loaded below this limit, materials do not fail (time considered by the graph)



Fatigue

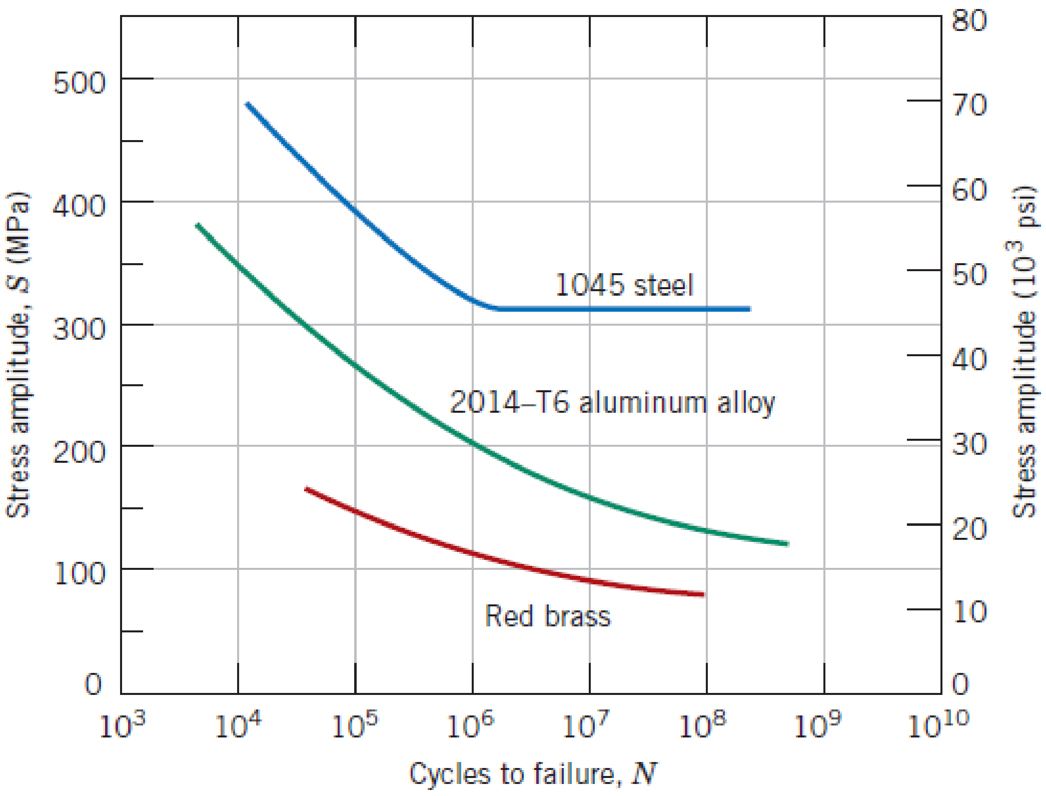
More cycles





5- A Brass cylinder (8 mm diameter) is cyclically loaded at +7500 N e -7500 N along its axis. Use curves above to calculate its fatigue life.

A steel rod is cyclically loaded at S= 22 kN along its axis. Use curves above to calculate the minimum diameter for this component to avoid fatigue fracture. Use a safety coefficient 2.



Es 5. The data obtained from a series of fatigue tests for a material are as follows:

σ (MPa)	N°
248	$1 \cdot 10^5$
236	$3 \cdot 10^5$
224	$1 \cdot 10^6$
213	$3 \cdot 10^6$
201	$1 \cdot 10^7$
193	$3 \cdot 10^7$
193	$1 \cdot 10^8$
193	$3 \cdot 10^8$

After constructing the Wöhler curve, determine:

a) The fatigue limit
b) The number of cycles to failure at 230 and 175 MPa
c) The fatigue strength at $2 \cdot 10^5$ and $6 \cdot 10^6$ cycles

6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: unfinished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.